The Monster Group $\mathbb{M}$

The largest of the 26 *sporadic simple groups*. To say what that means we have to back up.

What is a sporadic group?

Groups, briefly

A *group* is a set with a way to combine two elements (think "multiplication") that is associative, has an identity, and has inverses. Examples:

- the integers $\mathbb{Z}$ under addition,
- the symmetries of a square (rotations and reflections, 8 of them),
- the permutations of n objects — the symmetric group S_n, size $n!$.

A group is *finite* if it has finitely many elements. The "shape" of a finite group is captured by its multiplication table.

Normal subgroups and "simple"

Given a group G, certain subgroups $N \subseteq G$ are *normal*: roughly, they sit symmetrically inside G so that you can "divide them out" and form a smaller *quotient group* G/N. Normal subgroups are the analogue, for groups, of factors of an integer.

A group is **simple** if it has no nontrivial normal subgroup (no proper N with $1 < |N| < |G|$). Simple groups are the "primes" of group theory: every finite group can be broken apart into simple pieces, and those pieces are unique (Jordan–Hölder). So if you want to understand all finite groups, the first job is to list all the finite simple groups.

The classification

This list was finished around 2004, after roughly fifty years of work by hundreds of mathematicians, in tens of thousands of journal pages. The result — the *Classification of Finite Simple Groups* — says that every finite simple group is one of:

1. $\mathbb{Z}/p\mathbb{Z}$ for a prime p (cyclic groups of prime order, the only abelian simple groups);
2. A_n for $n \geq 5$ (the alternating groups: even permutations of n objects);
3. a group of *Lie type* (16 infinite families, built from matrix groups over finite fields — things like $\mathrm{PSL}_n(\mathbb{F}_q)$);
4. or one of **26 sporadic groups**, which fit into none of the above families.

The first three categories are infinite families parameterised by p, n, or (n, q) — regular, predictable, generated by a recipe. The 26 sporadics are the leftovers: isolated exceptions that the recipes do not produce. They were each discovered one at a time, often by a person who suspected a particular group "ought to exist" based on combinatorial or geometric constraints and then constructed it by hand.

Brief timeline (see detailed version below)

- **1861–1873.** Mathieu finds the first five.
- **1965.** Janko discovers J_1 after a 92-year gap.
- **1965–1976.** Twenty more sporadics arrive, mostly predicted from involution centralisers.
- **1980.** The Monster and J_4 are constructed; the list closes at 26.

The Happy Family and the Pariahs

Of the 26 sporadics, 20 of them (including the Monster itself) sit inside $\mathbb{M}$ as subquotients — you can find them by taking a subgroup of $\mathbb{M}$ and then quotienting. Robert Griess named these twenty the *Happy Family*. The remaining six — J_1, J_3, J_4, Ru, Ly, ON — have no such relationship to the Monster and are called the *pariahs*.

That the Monster contains most of the other sporadic groups *inside* it is part of why it is the natural climax of the story.

Detailed timeline: the 26 sporadics in order of discovery

What follows is the canonical chronology. Where a group was *predicted* from theoretical constraints before being *constructed* as a concrete object, both dates are given. The predictor and constructor are often different people; this was normal in the 1960s and 1970s.

Era I — The Mathieu groups (1861–1873)

- **1861.** Émile Mathieu introduces M_{12} and M_{24}, with M_{11} as a subgroup of M_{12}.
- **1873.** Mathieu completes the picture with M_{22} and M_{23}, sitting inside M_{24}.

These five groups (the *Mathieu groups*) sit on top of the *Steiner system* $S(5, 8, 24)$ and the binary *Golay code*. After Mathieu, the subject went silent for 92 years — people assumed there were no more sporadic groups, or did not know how to look.

Era II — The modern era opens (1965–1968)

- **1965.** Zvonimir Janko discovers J_1 by analysing a hypothetical involution centraliser of the form $\mathbb{Z}_2 \times A_5$. J_1 is a 7×7 matrix group over $\mathbb{F}_{11}$. It is the first new sporadic group in almost a century, and it convinces everyone that more must exist.
- **1967.** Donald Higman and Charles Sims, while attending a conference, construct the *Higman–Sims group HS* over a single weekend as a permutation group on 100 points.

- **1968.** Conway computes $\mathrm{Aut}(\Lambda_{24})$ and identifies Co_1, Co_2, Co_3. Three sporadics in one calculation.
- **1968.** Janko predicts a second group J_2 (= HJ, the *Hall–Janko group*); Marshall Hall constructs it as a rank-3 permutation group.
- **1968.** Michio Suzuki introduces the *Suzuki sporadic group Suz*, inside Co_1.
- **1968.** Jack McLaughlin introduces the *McLaughlin group McL*, related to the Leech lattice.
- **1968.** Dieter Held predicts the *Held group He*, with involution centraliser modelled on M_{24}. Constructed shortly after by Graham Higman and John McKay.

Era III — Janko's third, Fischer's family (1969–1972)

- **1969.** Janko predicts J_3. G. Higman and J. McKay construct it (a third Janko group, unrelated to J_2 in any nice way).
- **1971.** Bernd Fischer, studying groups generated by 3-*transpositions* (involutions whose pairwise products have order 1, 2, or 3), introduces three new sporadics: Fi_{22}, Fi_{23}, Fi'_{24}. The Fischer groups form a tower: $Fi_{22} \subset Fi_{23} \subset Fi'_{24}$.
- **1972.** Richard Lyons predicts the *Lyons group Ly* from an involution centraliser modelled on *McL*. Charles Sims constructs it in 1973 using a 111-dimensional representation over $\mathbb{F}_5$ — an early major use of computers in sporadic-group construction.

Era IV — Monster prediction and the final stretch (1973–1976)

- **1973.** Arunas Rudvalis predicts the *Rudvalis group Ru*. Conway and David Wales construct it later that year.
- **1973.** *Fischer and Griess independently predict the Baby Monster B and the Monster* $\mathbb{M}$, both much larger than anything previously contemplated.
- **1975.** Janko predicts J_4 from an exotic involution centraliser involving M_{22}.
- **1975–76.** Michael O'Nan introduces the *O'Nan group ON*, predicted from a hypothetical involution centraliser.
- **1976.** Koichiro Harada and Simon Norton introduce the *Harada–Norton group HN*, predicted as an involution-centraliser quotient inside the Monster.
- **1976.** John Thompson predicts the *Thompson group Th* from a hypothetical centraliser connected to the exceptional Lie group $E_8(\mathbb{F}_3)$. Smith and Thompson construct it shortly afterwards.

Era V — The list closes (1977–1980)

- **1977.** Jeff Leon and Charles Sims construct the Baby Monster B by computer, using a permutation representation on 13,571,955,000 points.
- **1980.** Robert Griess constructs the Monster $\mathbb{M}$ by hand, as the automorphism group of his 196,884-dimensional Griess algebra. He writes the construction up in *The Friendly Giant* (1982).

- **1980.** Simon Norton, Richard Parker, John Conway and David Benson construct J_4 using a 112-dimensional representation over $\mathbb{F}_2$, with substantial computer help.

That is all 26. After August 1980, every involution centraliser that arose in the ongoing classification effort matched a group already on the list. Nobody has ever produced a credible candidate for a 27th.

Size of the Monster

$$|\mathbb{M}| = 808\,017\,424\,794\,512\,875\,886\,459\,904\,961\,710\,757\,005\,754\,368\,000\,000\,000$$

which is about 8×10^{53}. The prime factorisation is

$$|\mathbb{M}| = 2^{46} \cdot 3^{20} \cdot 5^{9} \cdot 7^{6} \cdot 11^{2} \cdot 13^{3} \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71.$$

The 15 primes appearing here are the *supersingular primes*: exactly the primes p for which the modular curve $X_0(p)^+$ has genus zero. This coincidence has no business being true and was an early hint that something deep was happening.

History

- **1973.** Bernd Fischer and Robert Griess independently predict its existence from group-theoretic constraints.
- **1980.** Griess constructs $\mathbb{M}$ explicitly as the automorphism group of a 196,884-dimensional commutative non-associative algebra — the *Griess algebra*. His paper is titled "The Friendly Giant."

Smallest representations

- Trivial: dimension 1.
- Smallest faithful complex representation: dimension **196**,**883**.
- The Griess algebra: dimension 196,884 = 196,883 + 1.

Why exactly 26? Architecture of the classification proof

There is no single short proof. The classification of finite simple groups is a body of roughly 10,000–15,000 journal pages across about 500 papers by more than 100 mathematicians, spanning roughly 1955 to 2004. What follows is the *structure*: the load-bearing theorems and the strategy that ties them together.

Step 1. Reduce to groups with an involution

Feit–Thompson Odd Order Theorem (1963). Every finite group of odd order is solvable.

A solvable group is built from abelian pieces, and the only abelian simple groups are $\mathbb{Z}/p\mathbb{Z}$. So: *every nonabelian finite simple group has even order*, and therefore contains an element of order 2 (an *involution*). This is the foundation; the proof alone runs ≈ 255 pages and was a landmark.

After Feit–Thompson, the search reduces to nonabelian finite simple groups, all of which have involutions.

Step 2. Use the involution centraliser as a fingerprint

For $t \in G$ an involution, the *centraliser* $C_G(t) = \{g \in G : gt = tg\}$ records how t sits inside G.

Brauer–Fowler Theorem (1955). For any fixed finite group H, only finitely many finite simple groups G have an involution t with $C_G(t) \cong H$.

This is the theorem that makes the entire program possible. It says the involution centraliser is a near-fingerprint: pin it down, and the simple group is determined up to finitely many possibilities. The strategy of the classification is therefore:

1. enumerate the possible involution centralisers H that could occur in a nonabelian finite simple group;
2. for each H, identify the (finitely many) simple groups it produces.

Step 3. The big dichotomy

A nonabelian finite simple group G falls into one of two regimes, depending on the structure of $C_G(t)$:

- **Characteristic-2 type.** The centraliser of every involution has a generalised Fitting subgroup that is a 2-group. These groups behave like Chevalley groups in characteristic 2.
- **Component type.** Some involution centraliser has a quasi-simple "component." These groups behave like Chevalley groups in odd characteristic, or like alternating groups, or like sporadics.

The classification proof is, at the top level, a vast case analysis across this dichotomy. The two halves use different tools: Aschbacher's *component analysis* and the *signaliser functor method* (Gorenstein–Walter, Glauberman, Goldschmidt) on the component side; the *Bender method*, 2-local analysis, and weak-closure arguments on the characteristic-2 side.

Step 4. For each branch, match against a known list

In each leaf of the case analysis, one of these things must happen:

- $C_G(t)$ matches the involution centraliser of a known Chevalley group $\Rightarrow$ G *is* that Chevalley group (this is the bulk of the work and rules out enormous territory in one stroke);
- $C_G(t)$ matches the involution centraliser of an alternating group $\Rightarrow$ $G \cong A_n$;
- $C_G(t)$ matches the involution centraliser of a known sporadic $\Rightarrow$ G is that sporadic;
- $C_G(t)$ matches no known group $\Rightarrow$ a new sporadic is constructed and added to the list.

The last bullet is the only way new sporadics can enter. Each of the 26 was found exactly this way: someone was working through the case analysis, hit an involution centraliser that did not match

any known group, computed what such a group would have to look like (order, character table, key subgroups), and then constructed it. After J_4 was constructed in 1980, every branch terminated in an already-known group. No further branches remain.

Step 5. Verify the case analysis is exhaustive

This is the part that is genuinely $\sim$ 10,000 pages. The exhaustiveness is established in named subtheorems — e.g. Aschbacher's *classical involutions theorem* (1977), the *trichotomy theorem*, the *quasithin theorem*.

The gap, and the second-generation proof

In 1981 Gorenstein announced the classification was complete. In 1989 a gap was found in the *quasithin* case (groups of characteristic-2 type with small 2-local rank). The gap was not closed until 2004, when Aschbacher and Smith published a 1,221-page two-volume monograph filling it. With that, the classification is universally accepted.

A separate effort — the *second-generation proof*, started by Gorenstein, Lyons and Solomon in the 1990s — aims to give a more unified and streamlined account. It is roughly 13 volumes long and still in progress.

Why no 27th?

For a 27th sporadic group G to exist, every branch of the case analysis would already have to be wrong: some involution centraliser H would have to admit a simple group not on the list, and the branch of the proof that should have ruled this out would have to be incorrect. The two largest such risks — the quasithin case and the uniqueness theorems for the largest sporadics — have both been re-examined in detail. The proof is not aesthetically clean, but it is exhaustive.

This is why "find a new one" is not a research task one could pursue in good faith. It is the same shape of claim as "find a new prime between 7 and 11."

J_4: a pariah

Zvonimir Janko's fourth and final sporadic group, and one of the six *pariahs* — the sporadics that do *not* embed inside the Monster.

Order.

$$|J_4| \;=\; 2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43 \;\approx\; 8.68 \times 10^{19}.$$

Smallest faithful representation. Dimension 1,333 over $\mathbb{F}_{11}$ (and dimension 1,333 over $\mathbb{C}$ as well). The prime 11 plays a central structural role: $|J_4|$ is divisible by 11^3, and the group can only be seen comfortably in characteristic 11.

History. Predicted by Janko in 1975 on the basis of a hypothetical involution centraliser. The actual construction was a heroic effort by Simon Norton, Richard Parker, John Conway, David Benson and others, using a 112-dimensional representation over $\mathbb{F}_2$ and substantial computer work. It is the last sporadic group ever discovered: by the time J_4 was constructed in 1980, Griess had constructed the Monster, and the classification was complete.

Why pariah? The Monster's involution centralisers are double covers of B and of $2^{1+24}.Co_1$. Neither of these contains J_4, so J_4 has no home inside $\mathbb{M}$ — it is sporadic in the literal sense of standing apart, even from the rest of the family.

The Baby Monster B

The second-largest sporadic group, and the most natural companion to the Monster.

Order.

$$|B| \;=\; 2^{41} \cdot 3^{13} \cdot 5^6 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47 \;\approx\; 4.15 \times 10^{33}.$$

Roughly 10^{20} times smaller than $\mathbb{M}$, but still vastly bigger than any of the other 24 sporadics.

Smallest faithful representation. Dimension 4371 over $\mathbb{C}$. (Compare 196,883 for the Monster.) Like the Monster, B has no tiny faithful representation, just a much less extreme version of the same phenomenon.

History. Predicted by Bernd Fischer in the early 1970s — around the same time as the Monster — and constructed shortly after. Fischer called it the "baby monster" as shorthand while he was hunting for the larger group.

Relation to $\mathbb{M}$. Pick an involution $z \in \mathbb{M}$ of conjugacy class 2A (one of the two classes of order-2 elements in $\mathbb{M}$). Its centraliser is a double cover $2.B$, and quotienting by $\langle z \rangle$ recovers B itself. So the Baby Monster is, almost literally, "what you see when you look at $\mathbb{M}$ through a single involution."

Moonshine for B. Just as the Monster has Monstrous Moonshine attached to the j-function, B has its own *generalised moonshine* story: its character tables show up in the q-expansions of modular functions associated to particular elements of $\mathbb{M}$. The Baby Monster's moonshine is a corollary of the Monster's, but historically it was one of the early clues that the moonshine phenomenon was structural, not a coincidence restricted to a single group.

Monstrous Moonshine

In 1978, John McKay noticed:

$$196{,}884 \;=\; 196{,}883 + 1.$$

The left-hand side is a coefficient of the modular j-function,

$$j(\tau) \;=\; q^{-1} + 744 + 196{,}884\, q + 21{,}493{,}760\, q^2 + 864{,}299{,}970\, q^3 + \cdots$$

where $q = e^{2\pi i \tau}$. The right-hand side is a sum of dimensions of irreducible $\mathbb{M}$-representations. The pattern continues:

$$\begin{aligned} 21{,}493{,}760 &= 1 + 196{,}883 + 21{,}296{,}876, \\ 864{,}299{,}970 &= 2 \cdot 1 + 2 \cdot 196{,}883 + 21{,}296{,}876 + 842{,}609{,}326. \end{aligned}$$

Every coefficient of j is a small positive-integer combination of dimensions of Monster irreducible representations.

Conway and Norton formalised this as the *Monstrous Moonshine* conjectures: for each $g \in \mathbb{M}$, there is a modular function T_g whose q-expansion coefficients are characters of g acting on a graded $\mathbb{M}$-module $V^\natural$.

Borcherds proved this in 1992 (Fields Medal, 1998) by constructing $V^\natural$ — the *moonshine module* — as a vertex operator algebra, using generalised Kac–Moody algebras and ideas from string theory. The Monster turns out to be the symmetry group of a particular bosonic-string compactification on the Leech lattice modulo $\mathbb{Z}_2$.

Timeline of the classification proof

The proof of "there are exactly 26 sporadic groups" is really a proof of the larger theorem that classifies *all* finite simple groups. Sporadics are the leftovers, so finishing the classification finishes the count. Here are the milestones that matter, with each plainly explained.

- **1955 — Brauer–Fowler.** *Plain version:* if you tell me how an order-2 element sits inside a finite simple group (technically: its centraliser), I can narrow the group down to finitely many possibilities. This is what turns the classification from "infinite haystack" into "finite checklist."

- **1963 — Feit–Thompson.** *Plain version:* every nonabelian finite simple group has an order-2 element to work with. Without this you cannot even start Brauer–Fowler's strategy. The proof is 255 pages and was a milestone in its own right.

- **Late 1960s–early 1970s — the toolkit.** Mathematicians (Bender, Glauberman, Gorenstein, Walter, Aschbacher) build the machinery for analysing how order-2 elements interact: the *signaliser functor method*, 2-*local analysis*, *component analysis*. *Plain version:* each is a tool for turning a hypothetical centraliser into facts about the rest of the group.

- **1972 — Gorenstein's program.** Daniel Gorenstein lectures at the University of Chicago, laying out a 16-step roadmap for completing the classification. Up to this point, the field had been a chaotic land grab; afterwards, people knew which case to take.

- **1976 — Aschbacher's classical involutions theorem.** *Plain version:* a huge chunk of the checklist is closed in one paper. Groups with an involution whose centraliser "looks classical" are shown to be the already-known Lie-type groups. No surprises in that branch.

- **1976–1977 — the trichotomy theorem.** The remaining cases are split into three big buckets, each attacked separately. This is the structural skeleton of the rest of the proof.

- **1980 — the last sporadics are built.** Griess constructs the Monster. Norton, Parker, Conway and Benson construct J_4. After this, every open branch of the classification case analysis has been resolved by either a Lie-type group, an alternating group, or one of the 26 sporadics.

- **1981 — Gorenstein announces completion.** On the strength of preprints from many authors, Gorenstein declares the classification theorem proved.

- **1983 — the second-generation project.** Gorenstein, Richard Lyons and Ronald Solomon start writing a re-organised, streamlined proof. Roughly 13 planned volumes; about 10 have appeared so far.

- **1989 — the gap.** Michael Aschbacher and others realise that one branch of the case analysis — the *quasithin case* — was not actually fully proved. *Plain version:* "quasithin" is jargon for a specific narrow strip of the checklist (groups of "characteristic-2 type" with low "2-local rank"),

and the published proof for that strip had holes. Most people thought it would close quickly. It did not.

- **1992 — Gorenstein dies**, with the proof not yet definitively complete, though widely accepted.
- **2004 — Aschbacher and Smith close the gap.** Michael Aschbacher and Stephen Smith publish a two-volume, 1,221-page monograph that finally resolves the quasithin case. The classification of finite simple groups is now universally accepted as proved.
- **2008 onwards — the second-generation proof rolls on.** The Gorenstein–Lyons–Solomon series continues to appear, giving a more unified and shorter (still $\sim$ 5,000 pages) account.

How did they know J_4 was the last one?

In plain terms: by 1980, every branch of the giant case analysis had been resolved.

The strategy from 1955 onwards was a checklist. Each row of the checklist was a possible "shape" for the centraliser of an order-2 element in a finite simple group. Brauer–Fowler had guaranteed that each row corresponds to at most finitely many simple groups, so the checklist had a finite number of rows in principle, even if nobody knew the rows in advance.

For each row, only four outcomes are possible:

1. the row turns out to match a cyclic group $\mathbb{Z}/p\mathbb{Z}$ (the abelian simple groups);
2. the row matches an alternating group A_n;
3. the row matches a group of Lie type (e.g. $\mathrm{PSL}_n(\mathbb{F}_q)$);
4. the row matches none of the above, in which case a new sporadic group has to be constructed to fill it.

The four cases are exhaustive — this is the content of the classification theorem. Outcome (4) is the *only* way a new sporadic enters the list. Each of the 26 sporadics entered exactly this way.

By August 1980:

- Every row of the checklist had been worked through.
- Every row had been matched to one of the four cases.
- No row produced outcome (4) that was not already accounted for by one of the 26.

That is what "J_4 is the last one" actually means: there are no more rows of the checklist that produce a previously unknown group. The catch is that "every row of the checklist has been worked through" is itself a theorem, and theorems can have gaps — the quasithin gap above is the famous example. But since 2004, that gap is closed, and there are no other suspected holes.

So the question "could there be a 27th?" reduces to "could the proof be wrong?" That is not zero, but it is a different kind of question from "has anyone gone looking?".

John Conway and the problem

John Horton Conway (1937–2020) was a Cambridge lecturer with a reputation, by his late twenties, for cleverness without a flagship result. He was conscious of it. The Leech lattice gave him his problem, and once he had it he kept finding new ways back into the sporadic-group world for the next half-century.

The Leech lattice (1965)

In 1965, John Leech constructed an extraordinarily symmetric lattice Λ_{24} in 24-dimensional space, arising from the binary Golay code. It gives the densest known sphere packing in 24 dimensions (proved optimal only in 2017, by Viazovska and collaborators). Leech suspected its symmetry group $\mathrm{Aut}(\Lambda_{24})$ was enormous and probably a new sporadic group, but he could not pin down its order, and he could not find a group theorist willing to take the problem on.

Conway locks himself in (1968)

Conway took the bait. The story — which he told often, with small variations — is that he told his wife he would spend twelve hours every weekday and the full weekend on the problem, "for as long as it takes," starting on a Saturday. He emerged on the same Saturday night. The full structure had fallen out in roughly twelve hours.

He had computed

$$|\,\mathrm{Aut}(\Lambda_{24})| \;=\; |Co_0| \;=\; 2^{22}\cdot 3^{9}\cdot 5^{4}\cdot 7^{2}\cdot 11\cdot 13\cdot 23,$$

about 8.3×10^{18}, and identified three new sporadic simple groups in one stroke:

- $Co_1 = Co_0/\{\pm 1\}$;
- Co_2, the stabiliser of a type-2 vector in Λ_{24};
- Co_3, the stabiliser of a type-3 vector.

This is the only time in the history of the classification that multiple sporadics were uncovered in a single calculation by a single person. The paper, "A perfect group of order 8,315,553,613,086,720,000 and the sporadic simple groups" (1969), made his name.

The Monster, McKay, and moonshine (1973–1979)

- **1973.** Fischer and Griess independently predict the Monster from involution-centraliser arguments. Conway becomes one of the people computing what such a group would have to look like before anyone has constructed it.
- **1978.** John McKay, reading a number-theory paper on the train, notices $196{,}884 = 196{,}883 + 1$ — the first coefficient of the j-function written as a sum of Monster irreducible-representation dimensions. He mails Conway. Conway's first reaction, by his own account, is that it must be a coincidence. He checks the next coefficient, then the next, and very quickly stops believing in coincidence.
- **1979.** Conway and Simon Norton publish *Monstrous Moonshine*, formulating the precise conjectures: for each $g \in \mathbb{M}$ there is a specific genus-zero modular function T_g (a *Hauptmodul*) whose q-expansion coefficients are the characters of g on a hypothetical graded $\mathbb{M}$-module. They tabulate 194 such functions by hand.

- Conway coins the term "moonshine" for the phenomenon, in the sense of *something so wild it must be nonsense*. He kept the name on purpose, even after it stopped looking like nonsense.

Construction and aftermath (1980–1992)

- **1980.** Griess constructs $\mathbb{M}$ explicitly as the automorphism group of the 196,884-dimensional Griess algebra. In the same year, J_4 is constructed by a team including Conway, Parker, Norton and Benson; the classification's last sporadic falls into place.
- **1985.** The *ATLAS of Finite Groups* appears: Conway, Curtis, Norton, Parker and Wilson, a single oversized volume tabulating character tables, maximal subgroups and presentations for every finite simple group up to a tractable size, including all 26 sporadics. It is still the reference today.
- **1992.** Borcherds, building on work of Frenkel, Lepowsky and Meurman, proves the Conway–Norton conjectures by constructing the moonshine module $V^\natural$ as a vertex operator algebra. He is awarded the Fields Medal in 1998 for this work.

The rest of his life

Conway worked on sporadic groups, geometry of numbers, knot theory, and recreational mathematics until his death from COVID-19 on **11 April 2020**. He is more widely known for the Game of Life, the surreal numbers, the free will theorem, and the look-and-say sequence — but he repeatedly said that the Conway groups and moonshine were the work he was proudest of. The Monster, he liked to say, was the only group large enough to make him feel small.

Compact timeline

- **1937.** Conway born in Liverpool.
- **1964.** PhD, Cambridge (number theory, under Davenport).
- **1965.** Leech constructs Λ_{24}.
- **1968.** Conway computes $\mathrm{Aut}(\Lambda_{24})$ in a single weekend; Co_1, Co_2, Co_3 are born.
- **1973.** Fischer and Griess predict $\mathbb{M}$.
- **1978.** McKay's observation; Conway is convinced.
- **1979.** Conway–Norton, *Monstrous Moonshine*.
- **1980.** Griess constructs $\mathbb{M}$; J_4 constructed; list of sporadics closes at 26.
- **1985.** *ATLAS of Finite Groups* published.
- **1986.** Conway moves from Cambridge to Princeton.
- **1992.** Borcherds proves Monstrous Moonshine.
- **1998.** Borcherds, Fields Medal.
- **2004.** Aschbacher–Smith close the quasithin gap; classification accepted complete.
- **2020.** Conway dies, 11 April.

Why "monster"?

- It has no small faithful representation — 196,883 dimensions is the floor.
- It contains 19 of the other 26 sporadic groups as subquotients (the *Happy Family*). The remaining six are the *pariahs*: J_1, J_3, J_4, Ru, Ly, ON.
- Its connections to string theory, modular forms, lattice theory and the Leech lattice are real but still not fully understood.

References and further reading

For a non-specialist reader

- Mark Ronan, *Symmetry and the Monster: One of the Greatest Quests of Mathematics*, Oxford University Press, 2006. A readable narrative history of the entire classification, written for a general audience.
- Marcus du Sautoy, *Finding Moonshine: A Mathematician's Journey Through Symmetry*, Fourth Estate, 2008 (US title: *Symmetry*). A working mathematician's memoir-style tour of the subject.
- Terry Gannon, *Moonshine Beyond the Monster: The Bridge Connecting Algebra, Modular Forms and Physics*, Cambridge, 2006. Heavier going but the best single book on the moonshine side.

Survey and reference works

- Ronald Solomon, "A Brief History of the Classification of the Finite Simple Groups," *Bulletin of the AMS* **38** (2001), 315–352. The standard concise survey by one of the people who wrote the proof.
- J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, R. A. Wilson, *ATLAS of Finite Groups*, Clarendon Press, 1985. The reference work: character tables, maximal subgroups and presentations of all 26 sporadics and the small finite simple groups generally.
- D. Gorenstein, R. Lyons, R. Solomon, *The Classification of the Finite Simple Groups*, AMS Mathematical Surveys and Monographs 40, multi-volume from 1994 onwards. The second-generation proof.

Key original papers

- R. Brauer and K. A. Fowler, "On groups of even order," *Annals of Mathematics* **62** (1955), 565–583.
- W. Feit and J. G. Thompson, "Solvability of groups of odd order," *Pacific Journal of Mathematics* **13** (1963), 775–1029.
- J. H. Conway, "A perfect group of order 8,315,553,613,086,720,000 and the sporadic simple groups," *Bulletin of the London Mathematical Society* **1** (1969), 79–88.
- J. H. Conway and S. P. Norton, "Monstrous Moonshine," *Bulletin of the London Mathematical Society* **11** (1979), 308–339.

- R. L. Griess Jr., "The Friendly Giant," *Inventiones Mathematicae* **69** (1982), 1–102.
- R. E. Borcherds, "Monstrous moonshine and monstrous Lie superalgebras," *Inventiones Mathematicae* **109** (1992), 405–444.
- M. Aschbacher and S. D. Smith, *The Classification of Quasithin Groups*, AMS Mathematical Surveys and Monographs 111–112, 2004.

Lectures and talks

- Richard Borcherds has posted accessible lecture videos on moonshine and vertex algebras on YouTube.
- Conway gave many filmed lectures on the Monster and the Leech lattice; several are on YouTube under "John Conway Monster."

The mystery

We have a construction, and we have a proof, but we still lack a truly conceptual explanation for *why* the Monster exists, or why it is bound up with modular functions. Borcherds, on proving the moonshine conjectures: "I was over the moon …"